

# Grid-Integration of Battery Electric Vehicles

## Consumer Preferences for Smart Charging Programs

Pingfan Hu, Brian Tarroja,  
Matthew Dean, Kate Forrest,  
Eric Hittinger, Alan Jenn,  
John Paul Helveston



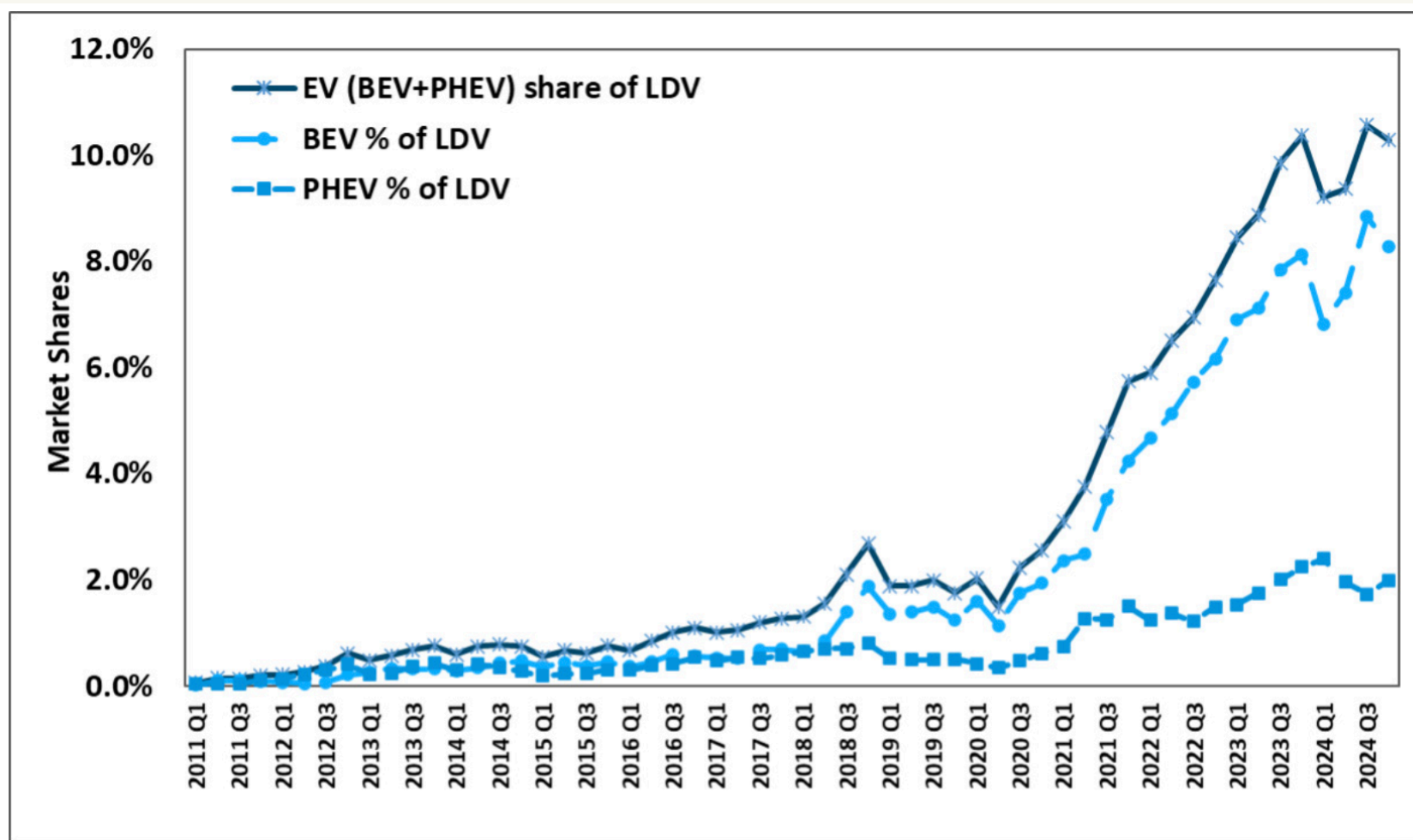
## About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
  1. **Data Science** in sustainable transportation and human behavior
  2. **Open-Source Software** in  tools for survey research
  3. **Agentic AI Engineering** around Claude Code
- For more information, visit [pingfanhu.com](https://pingfanhu.com)



# EV sales in US reaching ~10% of sales



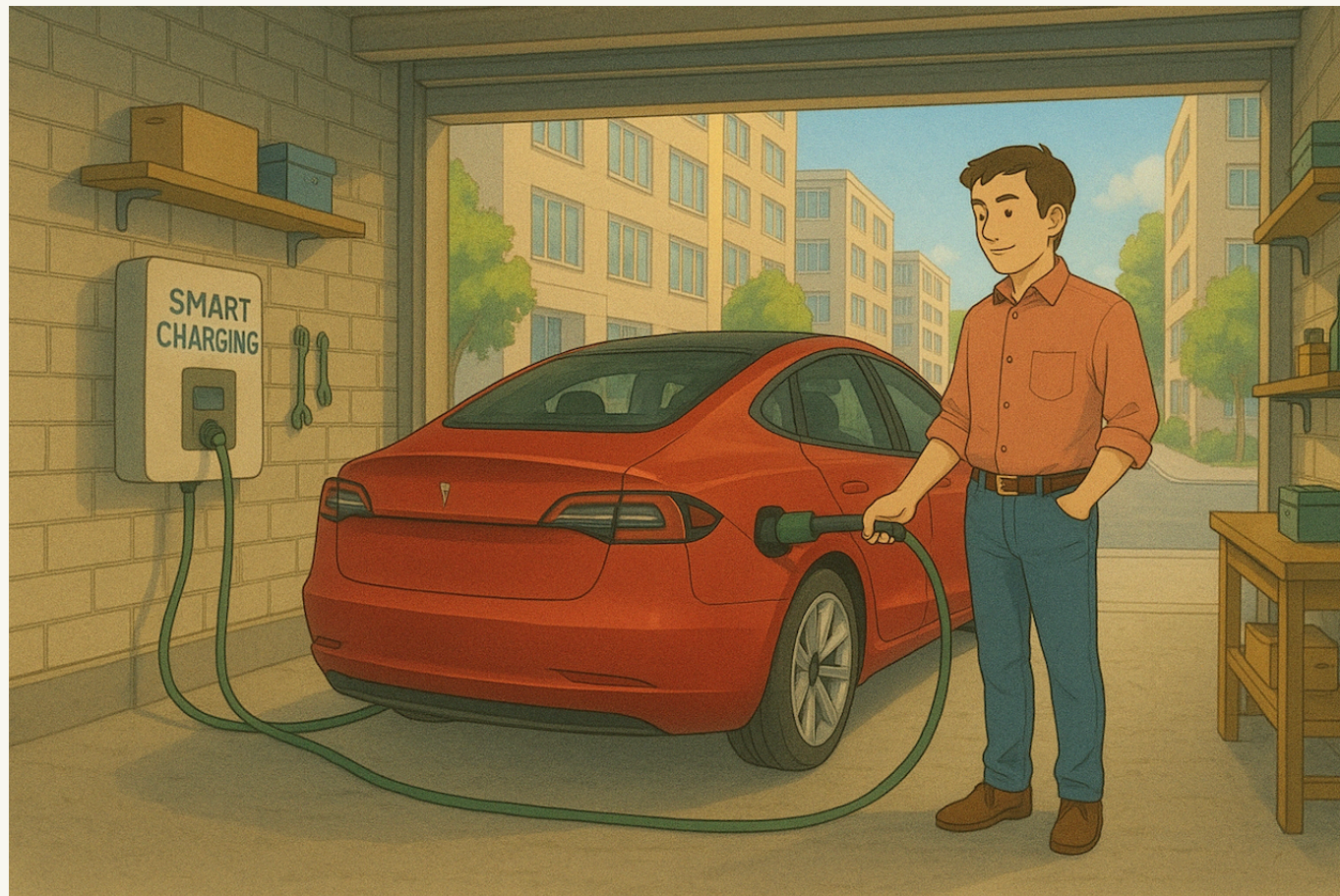
Source: Argonne National Lab, [www.anl.gov/ev-facts/model-sales](http://www.anl.gov/ev-facts/model-sales)





# Background

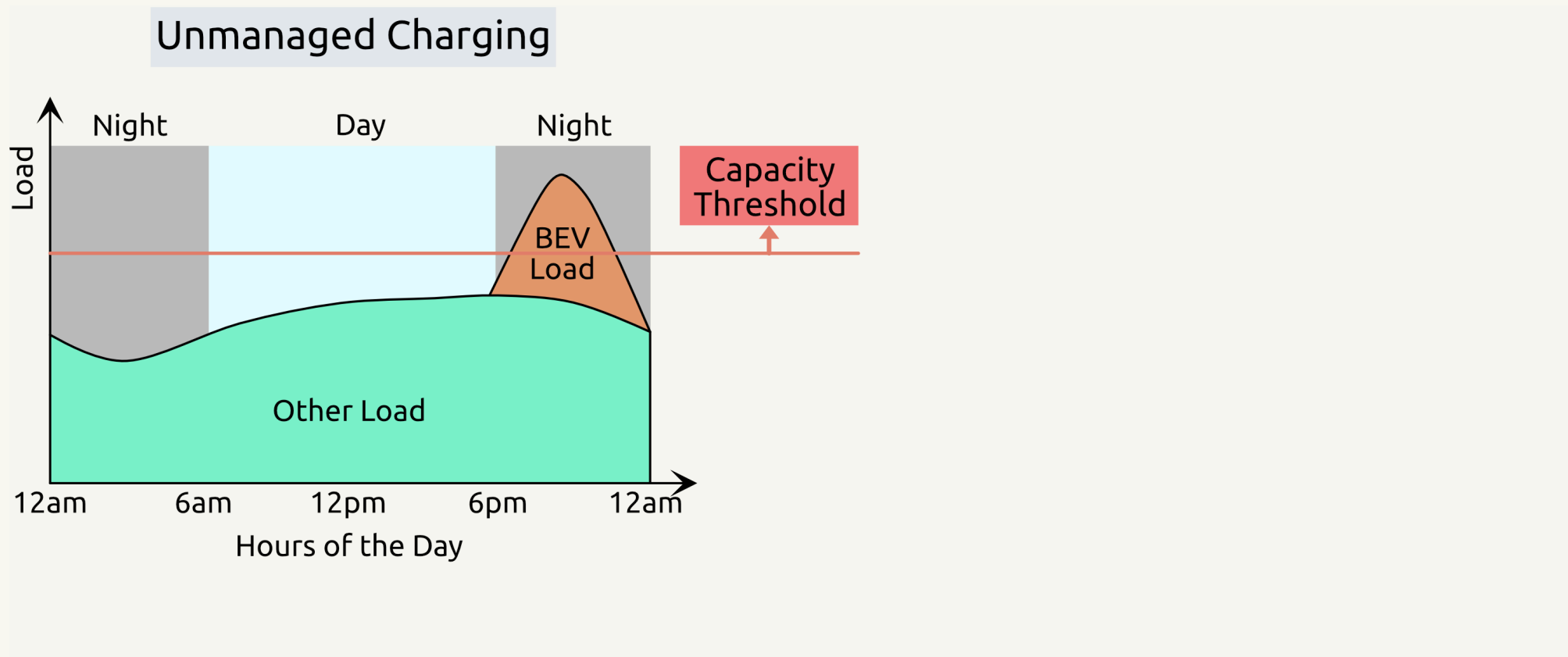
- **Unmanaged** BEV charging is becoming a problem to the grid.
- **Managed** charging is cheaper and smooths out the grid load.
- **Smart** charging: Supplier-Managed Charging (SMC) and Vehicle-to-Grid (V2G).





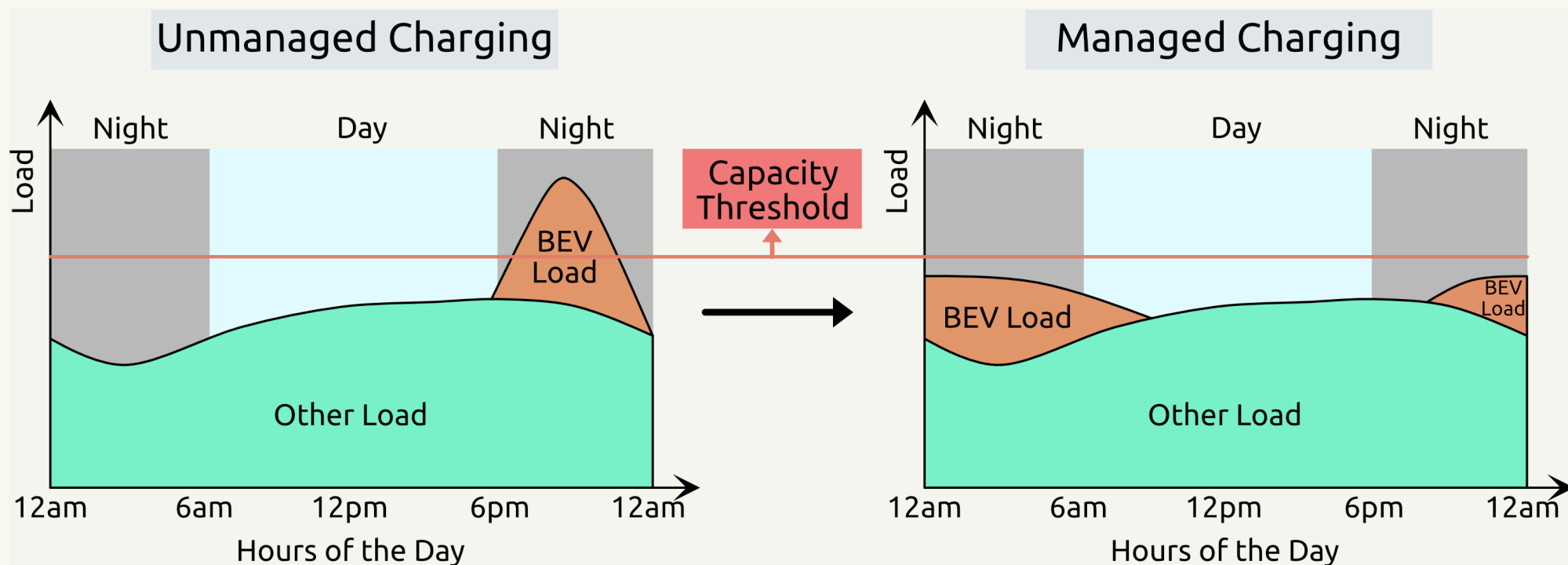
# SMC - Supplier Managed Charging

- SMC smooths out overnight BEV charging demand.
- Electricity demand is controlled below the capacity threshold.
- It saves money and reduces pollution.



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Managed charging avoids overload caused by BEV charging.





# V2G - Vehicle-to-Grid

## Non-V2G (Single Direction)

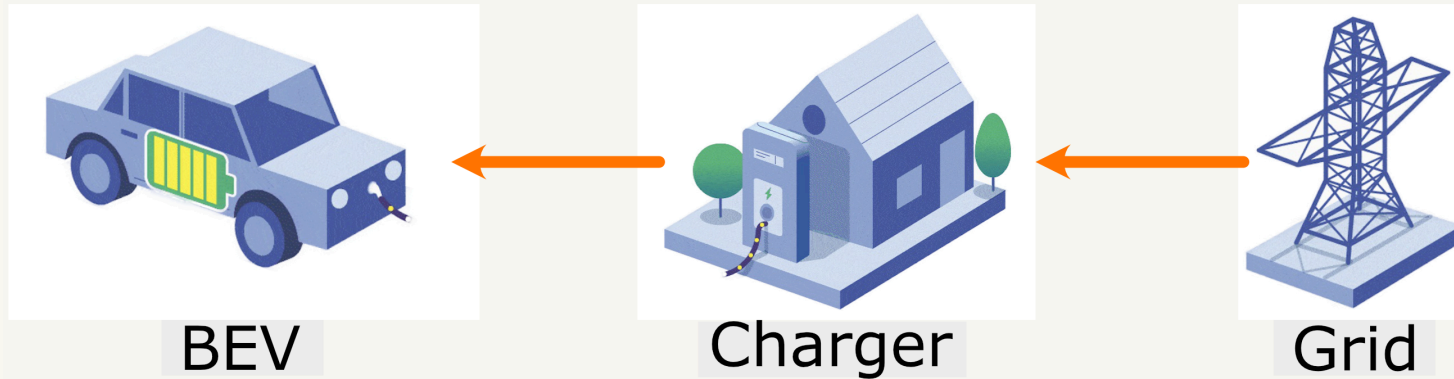
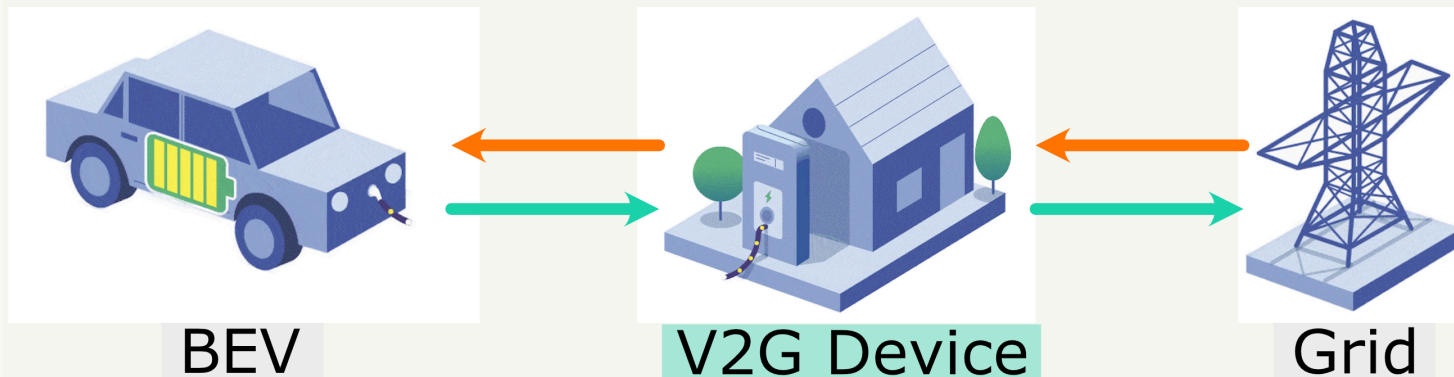


Figure Source: [wri.org](http://wri.org)

## V2G (Bi-direction)



In a V2G event, BEVs can charge the grid when necessary. BEVs are charged back eventually. Owners earn money.



**Smart charging depends on enrollment.**



# Literature Review

1. A study by Wong et al. (2023) examined how **incentives** affect EV owners' acceptance, **but EV ownership is only 19%**.
2. A study by Philip and Whitehead (2024) found **range anxiety** matters, **but EV ownership is only 1.28%**.
3. Another study by Huang et al. (2021) indicates the importance of **fast charging**, **but the sample size is only 157**.

None of them have demographic data to study **heterogeneity**.

**We need high BEV ownership & a large sample size, and consider heterogeneity.**





## Research Questions

1. **Sensitivity:** How do changes in smart charging program **features** influence BEV owners' willingness to opt in?
2. **Enrollment Rate:** Under what **combinations of features** will BEV owners be more willing to opt in to smart charging programs?

**Conjoint survey** to collect BEV owners' willingness.

**Multinomial logit model** for utility simulations.



# Survey Design with formr

## Conjoint Questions

1. Monetary Incentives
2. Charging Limitations
3. Flexibility

## Demographic Questions

1. BEV Ownership
2. Personal Info
3. Household Info



# Conjoint Question Explained

## A Sample Conjoint Question

For example, if these were the only apples available, which would you choose? \*

| Option 1                                                                          | Option 2                                                                            | Option 3                                                                            |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |
| <b>Type:</b> Fuji                                                                 | <b>Type:</b> Pink Lady                                                              | <b>Type:</b> Honeycrisp                                                             |
| <b>Price:</b> \$ 2 / lb                                                           | <b>Price:</b> \$ 1.5 / lb                                                           | <b>Price:</b> \$ 2 / lb                                                             |
| <b>Freshness:</b> Average                                                         | <b>Freshness:</b> Excellent                                                         | <b>Freshness:</b> Poor                                                              |

1. Provide respondents with different **sets** of attributes.
2. Observe choices across random sets.
3. Estimate **utility** of each attribute.

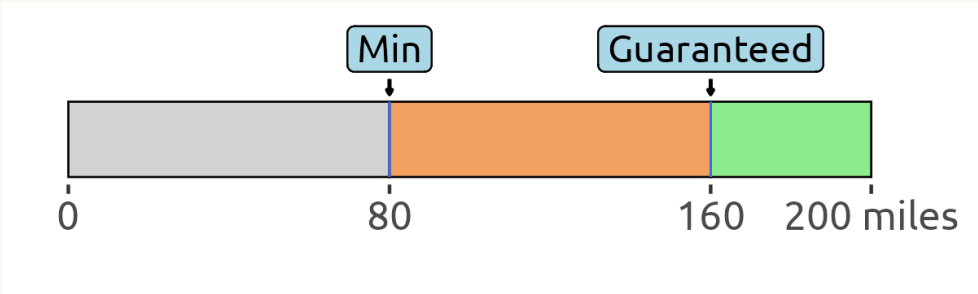


# SMC Programs

## Attributes

| No. | Attributes         | Range         |
|-----|--------------------|---------------|
| 1   | Enrollment Cash    | \$50 to \$300 |
| 2   | Monthly Cash       | \$2 to \$20   |
| 3   | Monthly Override   | 0 to 5        |
| 4   | Min Battery        | 20% to 40%    |
| 5   | Guaranteed Battery | 60% to 80%    |

## Sample Program

| Attributes                                                                                                                                           | Values |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Enrollment Cash                                                                                                                                      | \$300  |
| Monthly Cash                                                                                                                                         | \$20   |
| Monthly Override                                                                                                                                     | 5      |
| <div></div> <p>(Range determined by stated vehicle they own)</p> |        |





# V2G Programs

## Attributes

| No. | Attributes         | Range         |
|-----|--------------------|---------------|
| 1   | Enrollment Cash    | \$50 to \$300 |
| 2   | Occurrence Cash    | \$2 to \$20   |
| 3   | Monthly Occurrence | 1 to 4        |
| 4   | Lower Bound        | 20% to 40%    |
| 5   | Guaranteed Battery | 60% to 80%    |

## Sample Program

| Attributes         | Values |
|--------------------|--------|
| Enrollment Cash    | \$300  |
| Occurrence Cash    | \$20   |
| Monthly Occurrence | 1      |

0 80 160 200 miles

Low Guaranteed

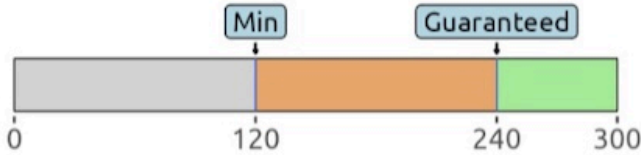
(Range determined by stated vehicle they own)



# Sample SMC Question

**(1 of 6)** If your utility offers you these 2 SMC programs, which one do you prefer?  
(Your BEV has maximum range of **300** miles.)

[Access the SMC Attributes](#)

| Option 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Option 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Option 3              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <p>Enrollment Cash: \$100</p> <p>Monthly Cash: \$20</p> <p>Override Allowance: 1 per Month</p> <p>Battery Thresholds (in Miles):</p>  <p>A horizontal bar representing a 300-mile range. It is divided into three segments: a grey segment from 0 to 60 miles, an orange segment from 60 to 240 miles, and a green segment from 240 to 300 miles. Above the bar, a blue box labeled 'Min' points to the 60-mile mark, and another blue box labeled 'Guaranteed' points to the 240-mile mark.</p> | <p>Enrollment Cash: \$200</p> <p>Monthly Cash: \$10</p> <p>Override Allowance: 1 per Month</p> <p>Battery Thresholds (in Miles):</p>  <p>A horizontal bar representing a 300-mile range. It is divided into three segments: a grey segment from 0 to 120 miles, an orange segment from 120 to 240 miles, and a green segment from 240 to 300 miles. Above the bar, a blue box labeled 'Min' points to the 120-mile mark, and another blue box labeled 'Guaranteed' points to the 240-mile mark.</p> | <p>Not Interested</p> |



# Sample V2G Question

**(1 of 6)** If your utility offers you these 2 V2G programs, which one do you prefer?  
(Your BEV has maximum range of **300** miles.)

[Access the V2G Attributes](#)

| Option 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Option 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Option 3              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <p>Enrollment Cash: \$100</p> <p>Occurrence Cash: \$5</p> <p>Monthly Occurrence: 2</p> <p>Battery Thresholds (in Miles):</p>  <p>A horizontal bar representing a 300-mile range. The bar is divided into three segments: grey (0-90 miles), orange (90-210 miles), and green (210-300 miles). A blue box labeled 'Low' points to the 90-mile mark, and a blue box labeled 'Guaranteed' points to the 210-mile mark.</p> | <p>Enrollment Cash: \$100</p> <p>Occurrence Cash: \$20</p> <p>Monthly Occurrence: 2</p> <p>Battery Thresholds (in Miles):</p>  <p>A horizontal bar representing a 300-mile range. The bar is divided into three segments: grey (0-120 miles), orange (120-180 miles), and green (180-300 miles). A blue box labeled 'Low' points to the 120-mile mark, and a blue box labeled 'Guaranteed' points to the 180-mile mark.</p> | <p>Not Interested</p> |



# Survey Fielding - 1,356 in Total



## Meta Ads: Voluntary participants

- 803 responses
- March to July in 2024

## Dynata Recruitment: Paid survey

- 553 responses
- September to November in 2024



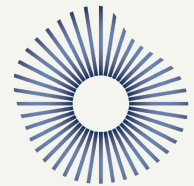
Facebook



Messenger



Instagram



**dynata**



# Survey Question - Car Ownership

## Car Ownership

1. What is your ZIP code?

22180

2. How many cars do you have?

- ☐ 1  
☒ 2  
☐ 3  
☐ 4  
☐ 5 or more

3. What is the **make** of your primary car?

3.1 What is the **model** of your primary car?

4. What is the **model year** of your primary car?

5. What is the **make** of your secondary car?

5.1 What is the **model** of your secondary car?

6. What is the **model year** of your secondary car?

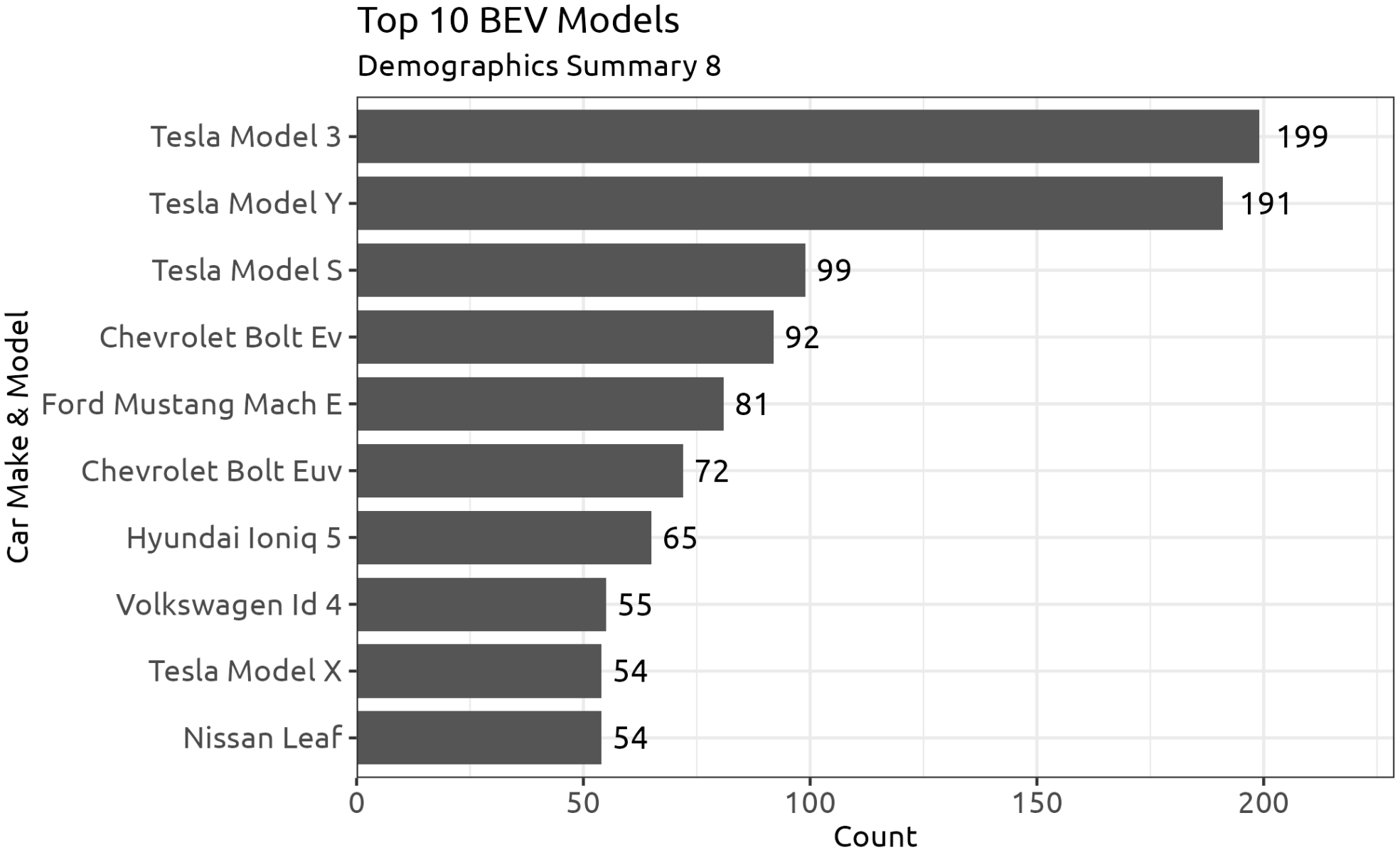
Other  
✓ Acura  
Alfa Romeo  
Aston Martin  
Audi  
Bentley  
BMW  
Bugatti  
Buick  
Cadillac  
Chevrolet  
Chrysler  
Daewoo  
Dodge  
Ferrari  
FIAT  
Fisker  
Ford  
Genesis  
GMC  
Honda

Next Page

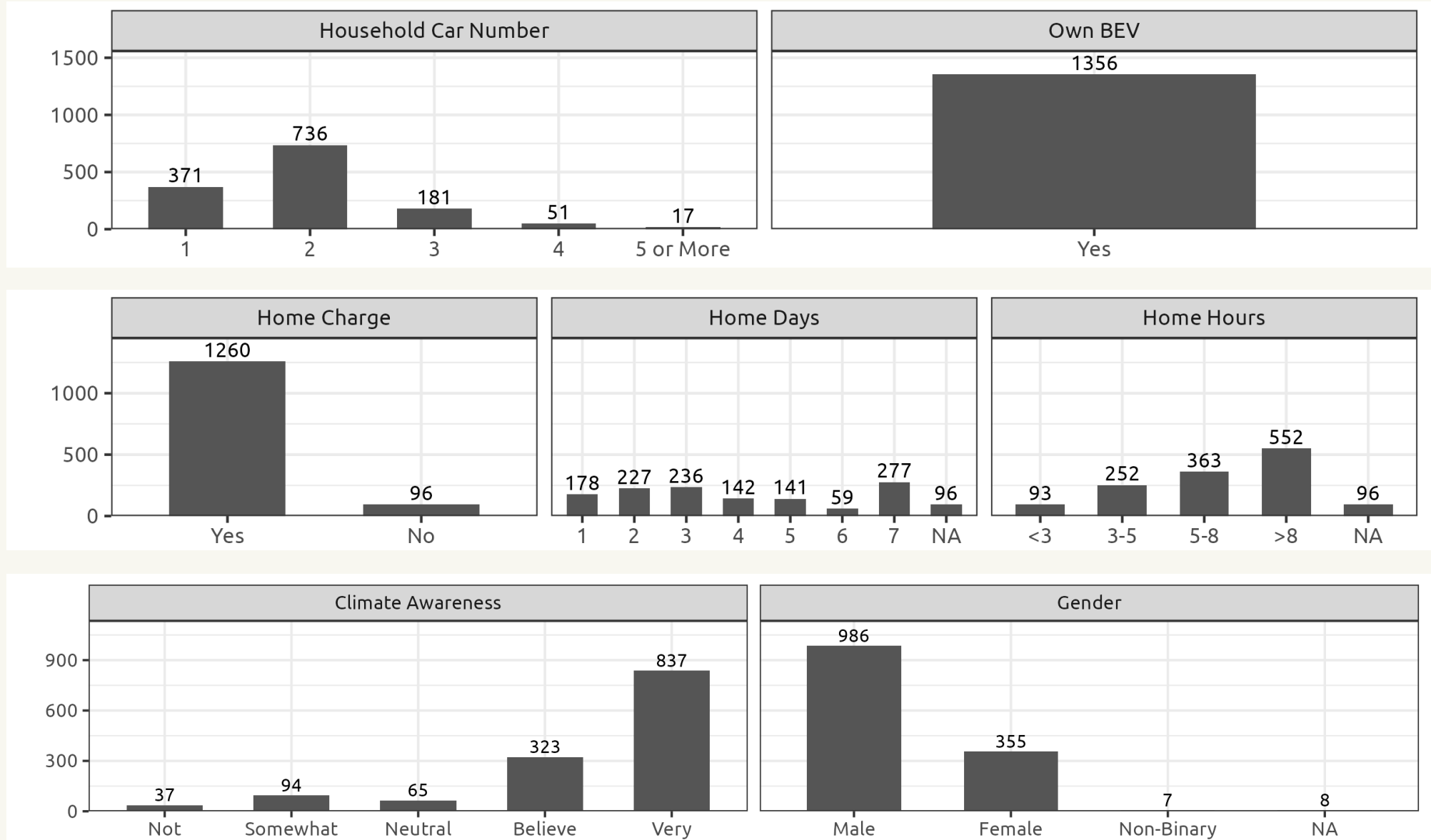




# Survey Results - Top 10 BEV



# Survey Results - Demographics



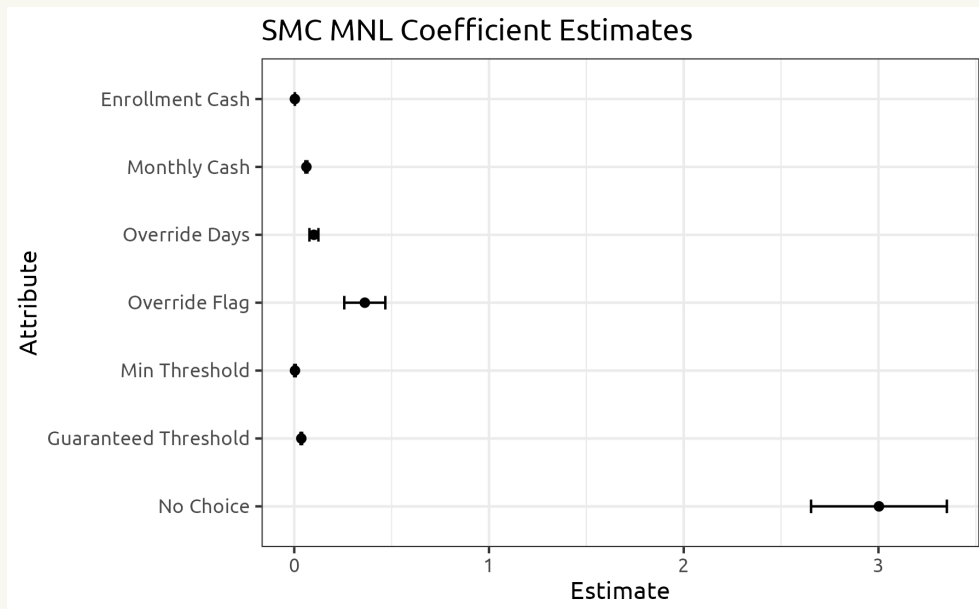
# Survey Results - Willingness to Participate

## Multinomial Logit Models

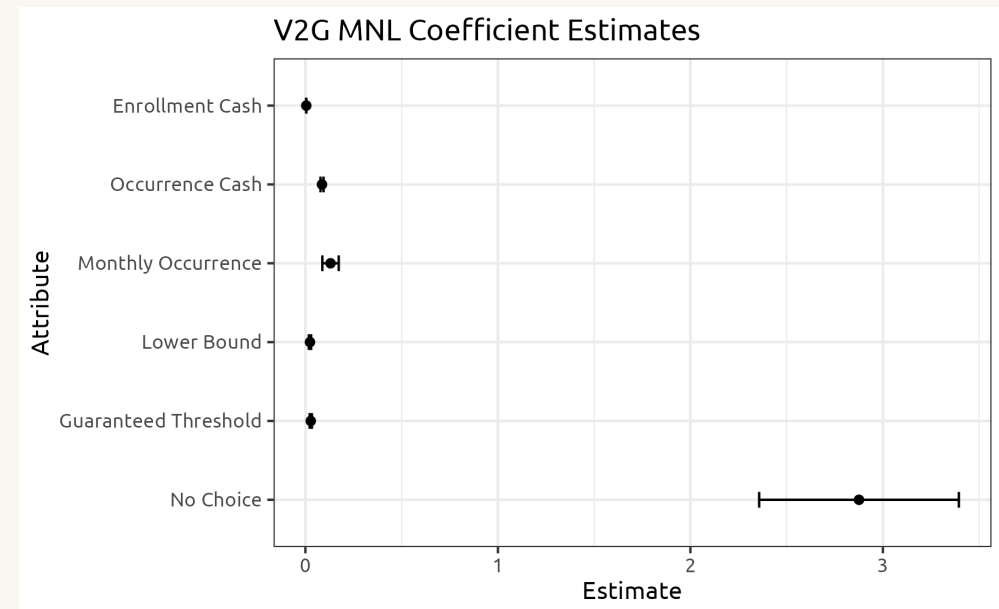
$$u_j = v_j + \epsilon_j = \beta' x + \epsilon_j \quad P_j = \frac{e^{v_j}}{\sum_{k=1}^J e^{v_k}}$$

Utility estimated using maximum likelihood estimation (MLE).

### SMC Estimates



### V2G Estimates



Without compensation, users will not participate.



# Enrollment Sensitivity

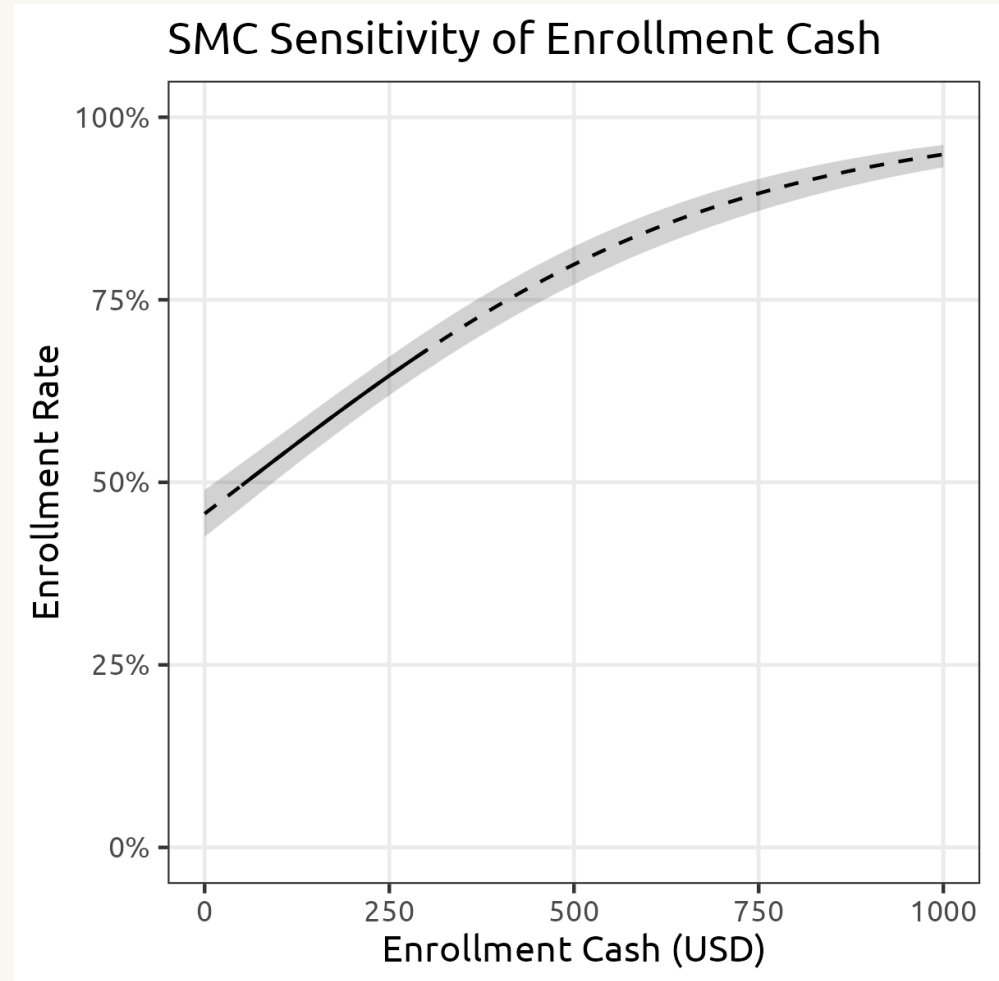
## Baseline Simulation

Choice between “None” and this program:

| Attributes       | Values       |
|------------------|--------------|
| Enrollment Cash  | \$0 - \$1000 |
| Monthly Cash     | \$2          |
| Monthly Override | 1            |

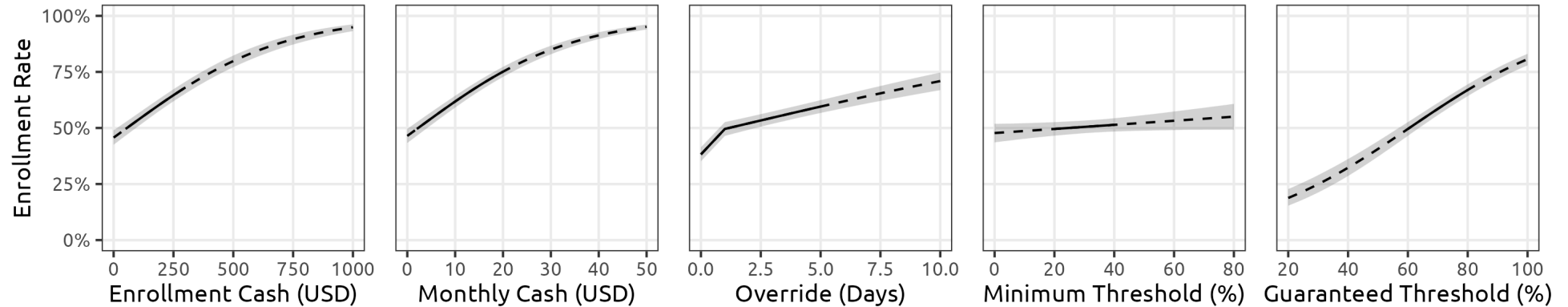
0 40 120 200 miles

## Sensitivity Plot

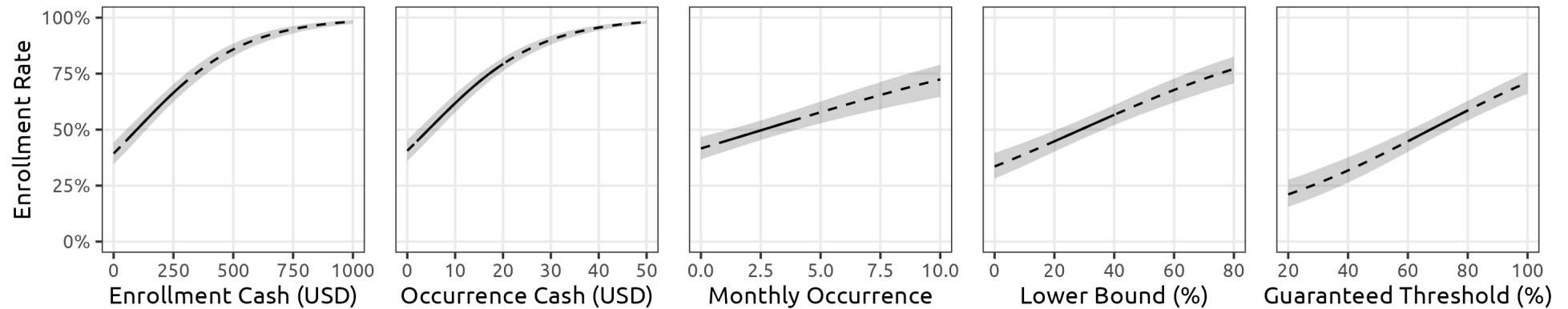


# Enrollment Sensitivity

## A) Supplier Managed Charging (SMC)



## B) Vehicle-to-Grid (V2G)



1. Steeper slope indicates higher sensitivity.
2. Diminishing returns exist.



# Equivalencies of 5% Enrollment Increase

## SMC

| Attribute                   | Equivalence Value | Unit     |
|-----------------------------|-------------------|----------|
| Enrollment Cash             | 77.7              | \$       |
| Monthly Cash                | 4.0               | \$       |
| Override Days               | 2.5               | Days     |
| Minimum Threshold           | 65.5              | %        |
| <b>Guaranteed Threshold</b> | <b>6.3</b>        | <b>%</b> |

## V2G

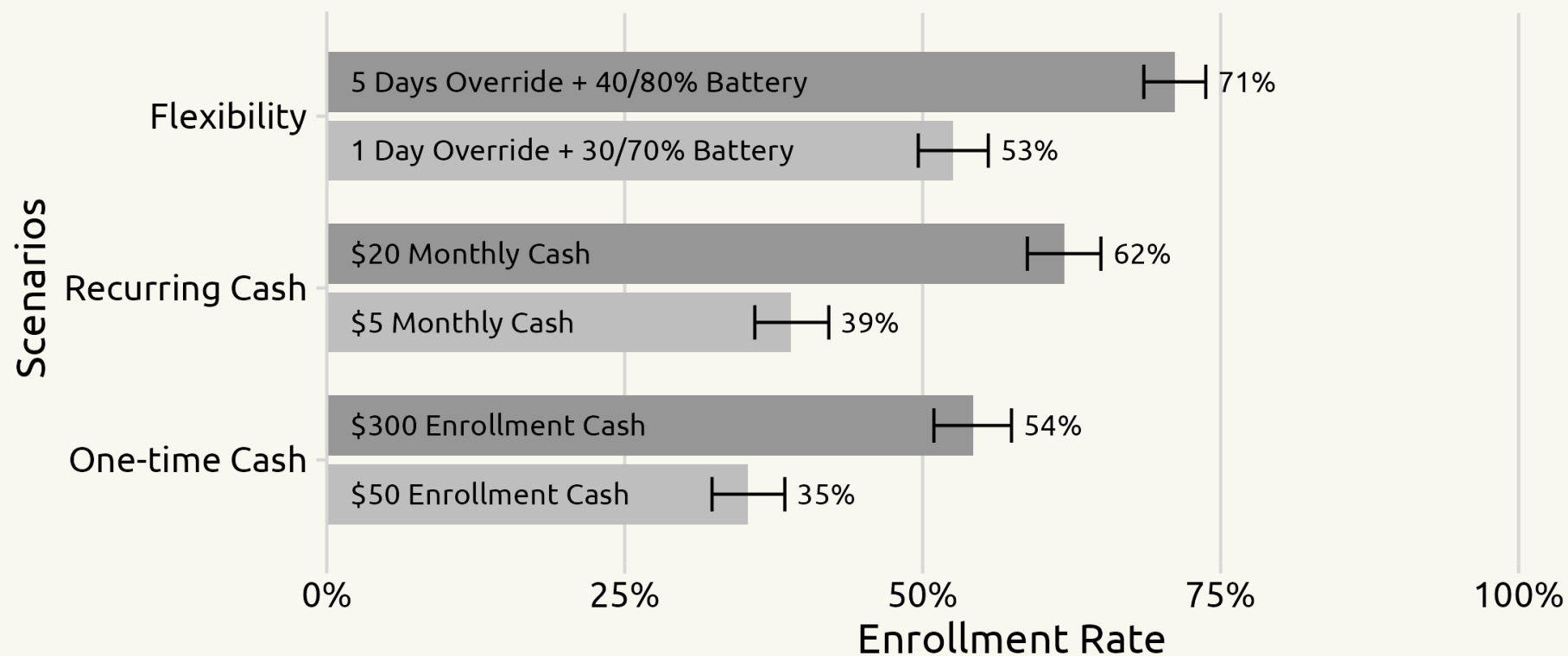
| Attribute              | Equivalence Value | Unit      |
|------------------------|-------------------|-----------|
| <b>Enrollment Cash</b> | <b>55.7</b>       | <b>\$</b> |
| <b>Occurrence Cash</b> | <b>2.9</b>        | <b>\$</b> |
| Monthly Occurrence     | 1.9               | Times     |
| Lower Bound            | 11.7              | %         |
| Guaranteed Threshold   | 9.1               | %         |

1. **Smaller** value indicates higher efficiency.
2. **Monetary** incentives are valued more in V2G than SMC.
3. **Guaranteed** thresholds are highly efficient, indicating range anxiety.
4. Attribute equivalencies can be used to inform incentive design.





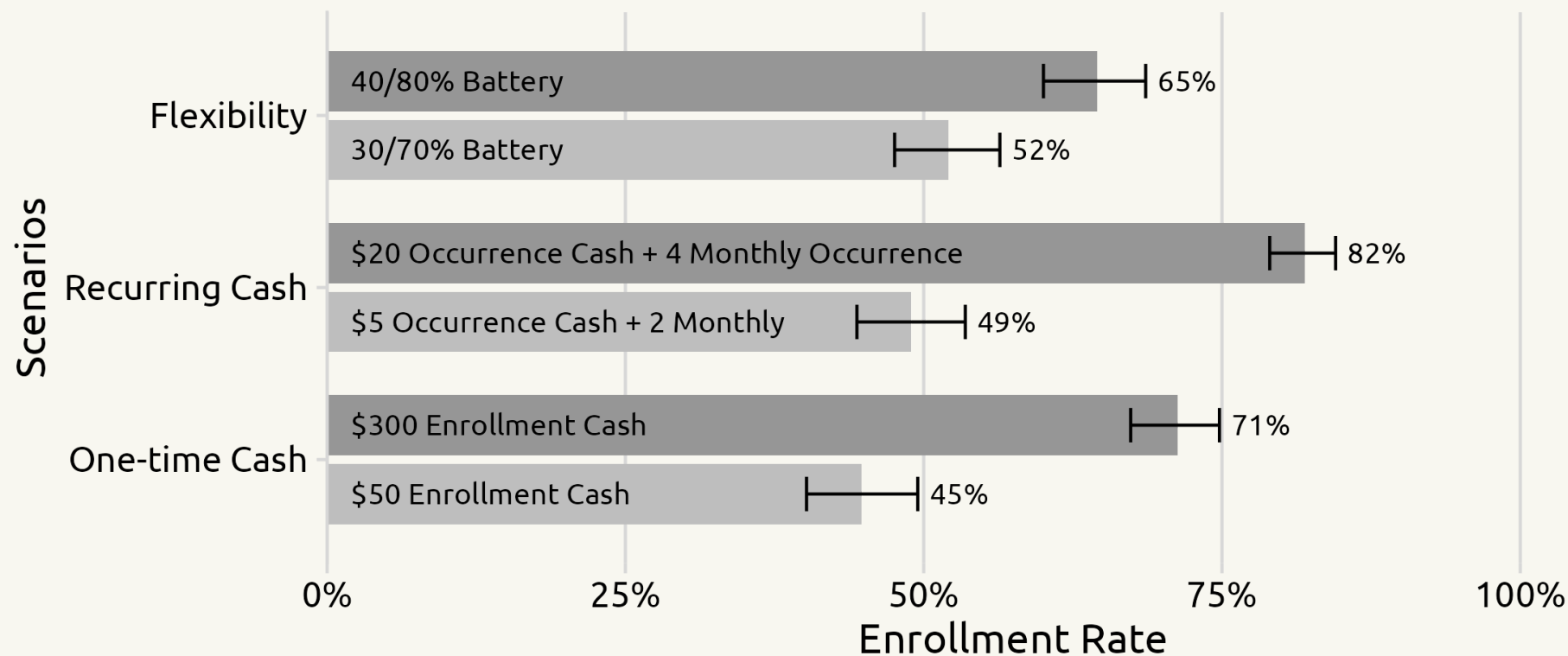
# SMC Scenario Analysis



1. **Flexibility** is highly valued.
2. **Recurring** incentives are more important than one-time ones.
3. Payment alone is not enough.



# V2G Scenario Analysis



1. Still, **recurring** incentives are more important than one-time ones.
2. But **flexibility** is not as important as in SMC.
3. Owners are willing to leverage BEV as a source of income.



# Smart Charging Enrollment Simulator

Smart Charging Enrollment Simulator
About
⚡ SMC (Supplier-Managed Charging)
🔌 V2G (Vehicle-to-Grid)

### SMC Attributes:

**Enrollment Cash (\$)**

**Monthly Cash (\$)**

**Override Allowance per Month**

**Minimum Threshold (%)**

**Guaranteed Threshold (%)**

Reset

### Predicted SMC Enrollment Probability:

## 31.9%

### About SMC:

- SMC (Supplier-Managed Charging) allows the utility to monitor, manage, and restrict BEV charging to optimize energy flow during night charging at home.
- By participating in SMC, your BEV will be mostly charged during off-peak periods.

### SMC Attributes Explained:

| Attribute                   | Description                                                                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enrollment Cash</b>      | The one-time payment you'll receive if you stay for at least 3 months.                                                              |
| <b>Monthly Cash</b>         | The recurring monthly payment you'll receive if you don't exceed override allowance.                                                |
| <b>Override Allowance</b>   | The monthly frequency of override to normal charging, effective for 24hrs. If you exceed the limit, no monthly cash for this month. |
| <b>Minimum Threshold</b>    | SMC won't be triggered below this threshold. In the survey it's converted to miles.                                                 |
| <b>Guaranteed Threshold</b> | SMC will give you this much of range by the morning (8 hrs' charging). In the survey it's converted to miles.                       |



## Contributions

1. First **large N** study of BEV owners' preferences for smart charging programs.
2. Quantified the **sensitivity** of BEV owners' preferences for smart charging features.
3. Introduced the concept of attribute **equivalencies** to inform incentive design.



## Appendix - SMC Logit Model

$$u_j = \beta_1 x_j^{\text{enroll\_cash}} + \beta_2 x_j^{\text{monthly\_cash}} + \beta_3 \delta_j^{\text{override\_allowed}} + \beta_4 x_j^{\text{num\_overrides}} \\ + \beta_5 x_j^{\text{min\_threshold}} + \beta_6 x_j^{\text{guaranteed\_threshold}} + \beta_7 \delta_j^{\text{no\_choice}} + \epsilon_j$$

| Attribute            | Coef.     | Est.   | SE     | Level             | Unit |
|----------------------|-----------|--------|--------|-------------------|------|
| Enrollment Cash      | $\beta_1$ | 0.0031 | 0.0002 | 50, 100, 200, 300 | USD  |
| Monthly Cash         | $\beta_2$ | 0.0623 | 0.0027 | 2, 5, 10, 15, 20  | USD  |
| Override Days        | $\beta_3$ | 0.1010 | 0.0118 | 0, 1, 3, 5        | Days |
| Override Flag        | $\beta_4$ | 0.3622 | 0.0538 | Yes, No           | -    |
| Minimum Threshold    | $\beta_5$ | 0.0037 | 0.0021 | 20, 30, 40        | %    |
| Guaranteed Threshold | $\beta_6$ | 0.0362 | 0.0021 | 60, 70, 80        | %    |
| No Choice            | $\beta_7$ | 3.0026 | 0.1779 | -                 | -    |



## Appendix - V2G Logit Model

$$u_j = \beta_1 x_j^{\text{enroll\_cash}} + \beta_2 x_j^{\text{occur\_cash}} + \beta_3 x_j^{\text{num\_occurrences}} + \beta_4 x_j^{\text{lower\_threshold}} \\ + \beta_5 x_j^{\text{guaranteed\_threshold}} + \beta_6 \delta_j^{\text{no\_choice}} + \epsilon_j$$

| Attribute            | Coef.     | Est.   | SE     | Level             | Unit  |
|----------------------|-----------|--------|--------|-------------------|-------|
| Enrollment Cash      | $\beta_1$ | 0.0045 | 0.0026 | 50, 100, 200, 300 | USD   |
| Occurrence Cash      | $\beta_2$ | 0.0863 | 0.0040 | 2, 5, 10, 15, 20  | USD   |
| Monthly Occurrence   | $\beta_3$ | 0.1305 | 0.0217 | 1, 2, 3, 4        | Times |
| Lower Threshold      | $\beta_4$ | 0.0237 | 0.0030 | 20, 30, 40        | %     |
| Guaranteed Threshold | $\beta_5$ | 0.0278 | 0.0030 | 60, 70, 80        | %     |
| No Choice            | $\beta_6$ | 2.8759 | 0.2647 | -                 | -     |



# Reference List

- Huang, Bing, Aart Gerard Meijssen, Jan Anne Annema, and Zofia Lukszo. 2021. “Are Electric Vehicle Drivers Willing to Participate in Vehicle-to-Grid Contracts? A Context-Dependent Stated Choice Experiment.” *Energy Policy* 156 (September): 112410. <https://doi.org/10.1016/j.enpol.2021.112410>.
- Philip, Thara, and Jake Whitehead. 2024. *Consumer Preferences Towards Electric Vehicle Smart Charging Program Attributes: A Stated Preference Study*. Rochester, NY. <https://doi.org/10.2139/ssrn.4812923>.
- Wong, Stephen D., Susan A. Shaheen, Elliot Martin, and Robert Uyeki. 2023. “Do Incentives Make a Difference? Understanding Smart Charging Program Adoption for Electric Vehicles.” *Transportation Research Part C: Emerging Technologies* 151 (June): 104123. <https://doi.org/10.1016/j.trc.2023.104123>.

